Urban Traffic Collision & Severity Analysis

Optimizing Patrol Deployment & Public Safety Through Data-Driven Analysis

Team ModelCraft | Newton School of Technology, Pune

Data Visualization & Analytics — Capstone 2 | April 2026

Team ModelCraft

Newton School of Technology, Pune — DVA Capstone 2

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Problem Statement

The Challenge

Despite traffic regulations, road accidents continue to cause significant casualties. Authorities lack granular understanding of how environmental conditions, driver demographics, and road surfaces interact to escalate minor collisions into fatal or serious accidents — making resource allocation reactive rather than preventive.

Core Business Question

How can city traffic departments optimize patrol deployments and targeted public awareness campaigns to proactively reduce the frequency and severity of road traffic accidents?

Decision Supported

Enable traffic planning departments to strategically allocate patrol resources during high-risk windows, prioritize infrastructure upgrades at dangerous junctions, and design targeted safe-driving initiatives for the highest-risk demographic and environmental segments.

Dataset Overview

12,316	32	3	27.9%
Records	Columns	Severity Levels	Adverse Conditions

Source: Kaggle — Road Traffic Accident Dataset | Format: CSV | Granularity: One row per accident record

Column	Description	Analytical Role
Accident_severity	Slight Injury / Serious Injury / Fatal injury	Target variable — drives all severity KPIs
Hour_of_day / Day_of_week	Temporal markers derived from Time	Temporal risk windows for patrol scheduling
Patrol_Shift	Morning / Evening / Night (derived)	Resource allocation across operational shifts
Road_surface_conditions	Dry / Wet or damp / Snow	Environmental risk factor segmentation
Age_band_of_driver	Under 18 / 18-30 / 31-50 / Over 51	Demographic targeting for safety campaigns
Severity_score	1 (Slight) / 2 (Serious) / 3 (Fatal)	Engineered numeric target for aggregation

4 derived columns engineered during ETL: Hour_of_day, Patrol_Shift, Is_Adverse_Condition, Is_weekend, Severity_score

ETL Pipeline & Data Cleaning

1

Extract

Raw CSV loaded from

data/raw/RTA_Dataset.csv

12,316 rows × 32 cols

2

Transform

Missing values imputed

Types standardized

5 derived columns created

3

Load

Clean dataset exported to

data/processed/RTA_cleaned.csv

Ready for Tableau & EDA

Key Transformations

- Handled missing values with mode imputation for categoricals and median for numerics
- Standardized inconsistent severity labels (e.g. 'Fatal injury' vs casing variants)
- Extracted Hour_of_day from timestamp and mapped to Patrol_Shift bins
- Created Is_Adverse_Condition flag from weather + light + road surface combination
- Engineered Severity_score (1/2/3) from Accident_severity for numeric aggregation
- Added Is_weekend binary flag for temporal segmentation

Pipeline: notebooks/01_extraction.ipynb → 02_cleaning.ipynb → scripts/etl_pipeline.py

KPI Framework

12,316

Total Accidents

All recorded incidents

1,901

Severe Accidents

158 Fatal + 1,743 Serious

15.4%

Severity Rate

(Fatal + Serious) / Total

19,067

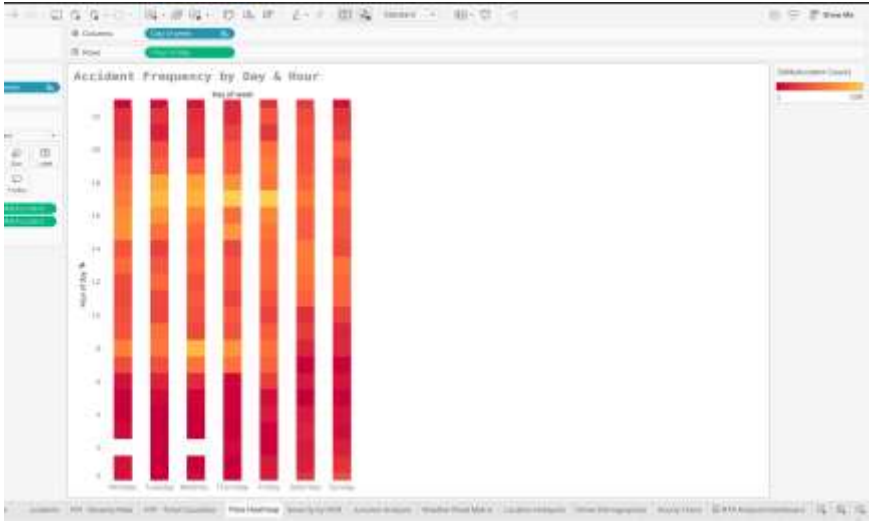
Total Casualties

Sum across all incidents

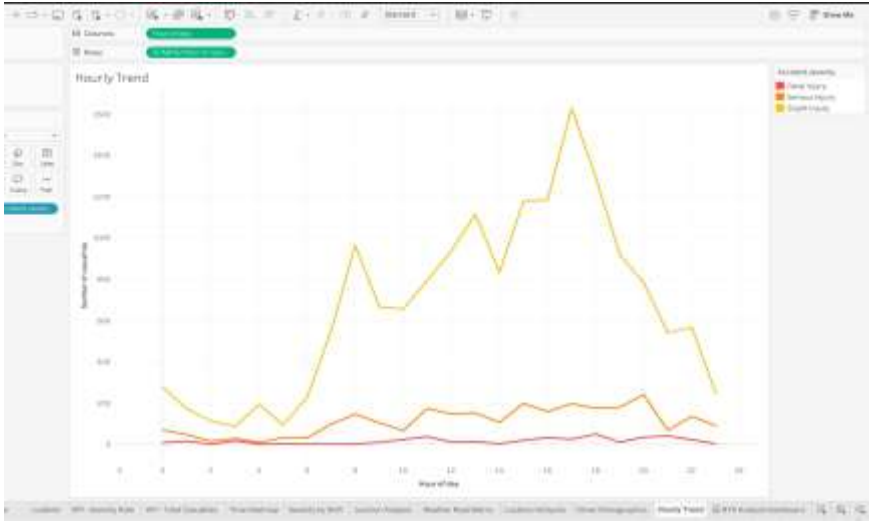
KPI	Formula / Computation	Decision Purpose
Severity Ratio	(Serious + Fatal) / Total Accidents = 1,901 / 12,316	Benchmark overall risk intensity
High-Risk Time Frequency	COUNT grouped by Hour_of_day & Day_of_week	Identify peak patrol windows
Adverse Condition Rate	27.9% — 3,441 under non-optimal conditions	Quantify environmental risk contribution
Shift Severity Index (LOD)	{ FIXED [Patrol_Shift] : AVG([Severity_score]) }	Compare baseline risk across shifts

Advanced: Dynamic metric selector parameter (CASE expression) + Above/Below Average LOD comparison used in dashboard tooltips

EDA: Temporal Patterns



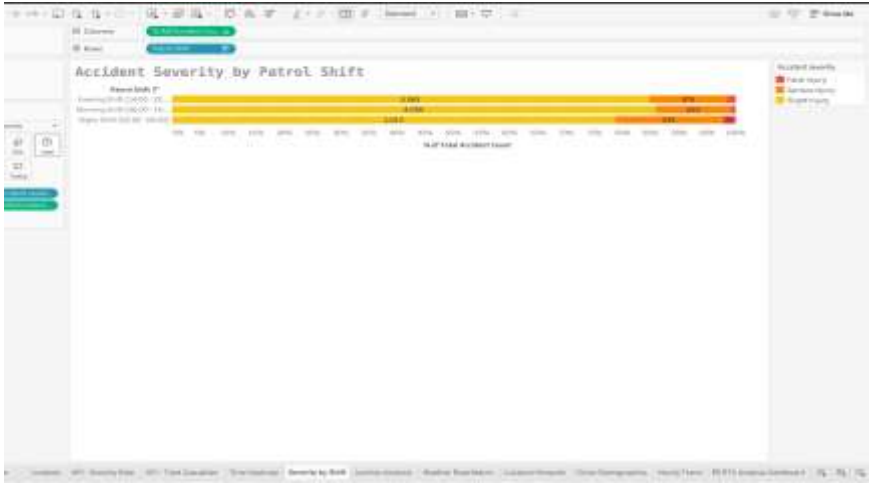
Time Heatmap (Day × Hour)



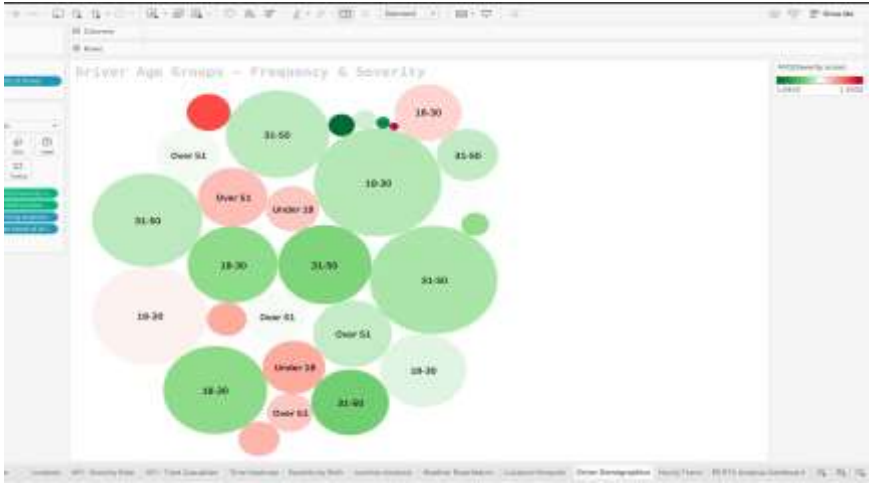
Hourly Trend by Severity

- Peak hour 17:00 alone accounts for 1,228 accidents (10% of total)
- 15:00–19:00 window = 38.1% of all accidents (4,687 incidents)
- Friday is the riskiest day (2,041) — 39% more than Sunday (1,467)

EDA: Shift, Severity & Demographics



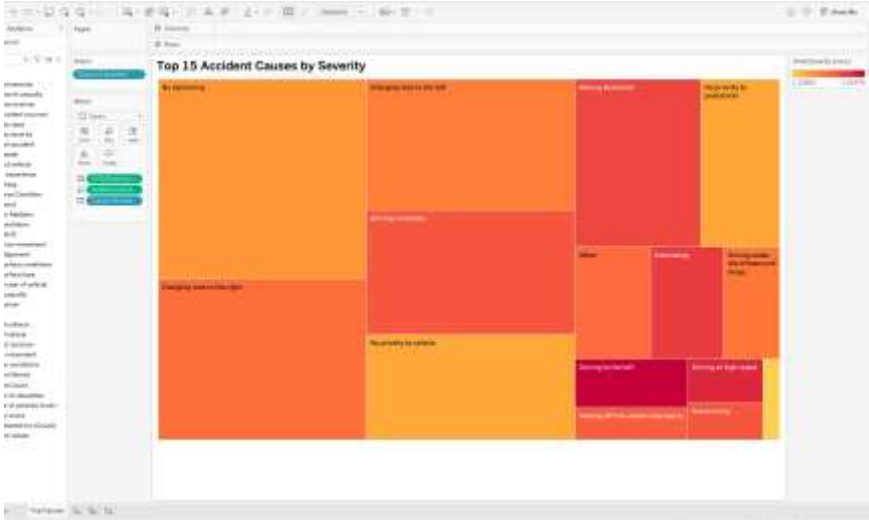
Severity by Patrol Shift



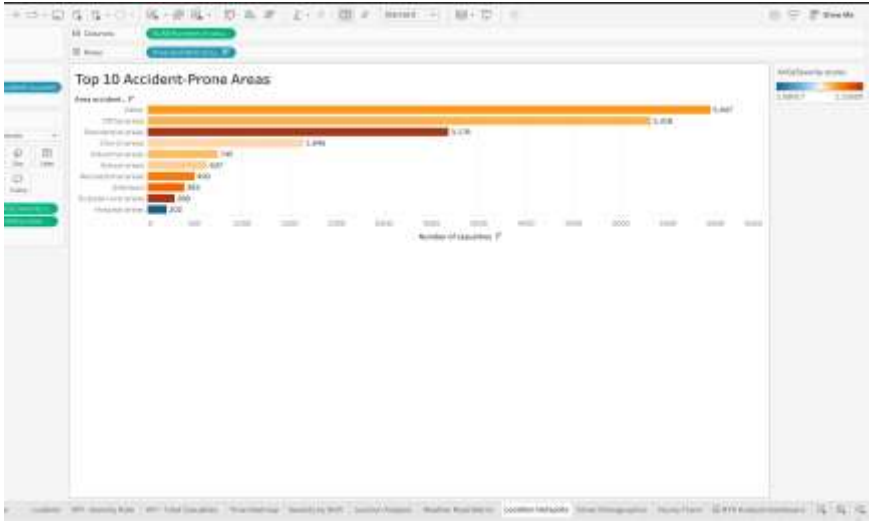
Driver Age Groups — Frequency & Severity

- Evening shift handles 51.4% of all accidents (6,331) with 92 fatalities — highest of any shift
- Night shift has the highest average severity score (1.235) despite lowest volume — severity per incident is elevated
- 18–30 age band: 34.7% of incidents (4,271) — the single largest demographic risk group
- Males account for 92.9% of all driver-involved accidents (11,437 of 12,316)

EDA: Causes, Locations & Environment



Top 15 Causes (Treemap)



Top 10 Accident-Prone Areas

- No distancing is the #1 cause (2,263 — 18.4%) — not overspeed. Tailgating interventions should be the top priority.
- Office areas (3,451) + Residential (2,060) = 44.7% of identifiable-area accidents — urban commute corridors are the primary risk zones
- 28.6% of accidents occur in darkness; average severity in darkness (1.19) exceeds daylight (1.16) — lighting infrastructure matters
- Y-Shape junctions are the most accident-prone (4,543 incidents) — 36.9% of all junction-related accidents

Statistical Analysis

Chi-Square Test: Severity × Patrol Shift

Categorical independence test on crosstab of Accident_severity × Patrol_Shift

Statistically significant relationship between patrol shift and accident severity. Night shift shows the highest average severity (1.235) despite handling only 10.4% of total volume — severity is not proportional to volume.

Correlation Analysis: Environmental Factors

Severity_score aggregated across Weather × Road_surface interaction matrix

Road surface condition and weather jointly influence severity. Normal weather + dry roads = baseline 1.17 avg severity, while rain + dry roads (counterintuitively) spikes to 1.29 — suggesting early-rain conditions catch drivers off-guard.

Distribution Analysis: Temporal Clustering

Hourly frequency distribution with density overlay and reference-line comparison

Accident frequency is bimodal — peaks at 08:00 (morning commute: 828) and 17:00 (evening commute: 1,228). Evening peak is 48% higher than morning, confirming asymmetric risk.

LOD Expression: Shift Severity Baseline

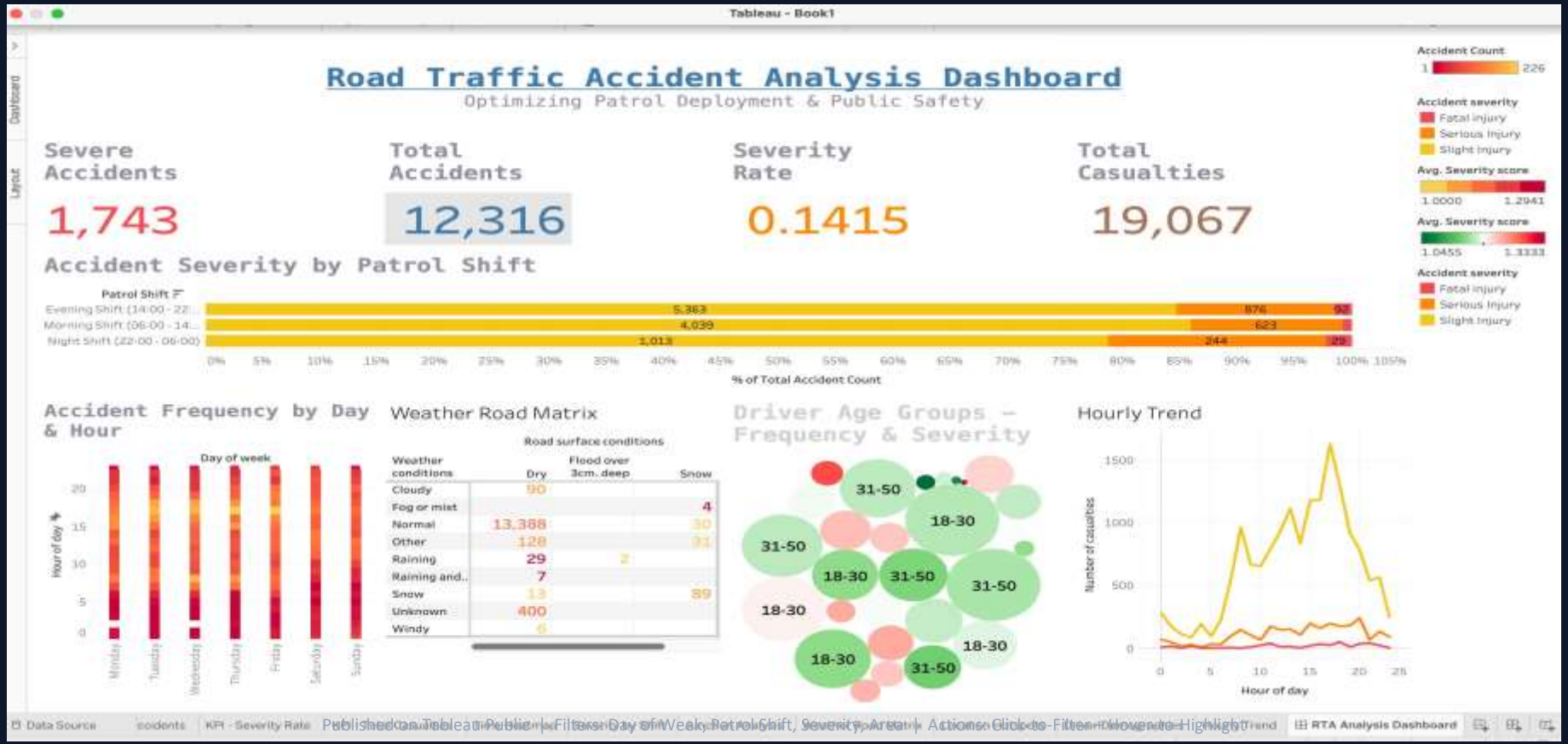
{ FIXED [Patrol_Shift] : AVG([Severity_score]) } — filter-independent baseline

FIXED LOD reveals Night shift average (1.235) exceeds Evening (1.167) and Morning (1.148). Used for per-record above/below-average classification and dashboard tooltips.

Detailed computations: notebooks/03_eda.ipynb & notebooks/04_statistical_analysis.ipynb

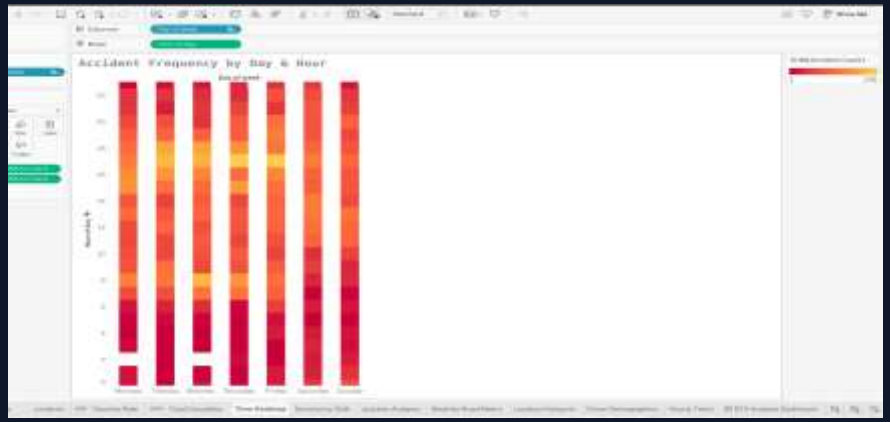
Tableau Dashboard

12 sheets | 4 cross-filters | LOD expressions | Dynamic metric parameter



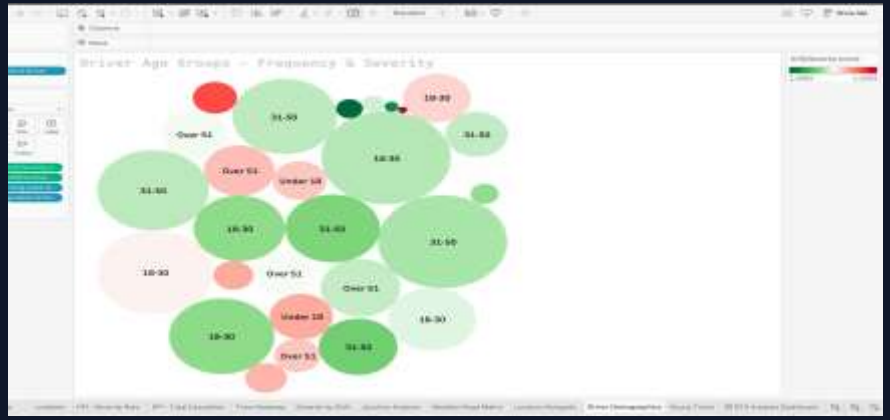
Published on Tableau Public | Filters: Day of Week, Patrol Shift, Severity, Area | Actions: Click-to-Filter, Hover-to-Highlight

Dashboard — Key Views



Time Heatmap (Day × Hour)

Severity by Patrol Shift

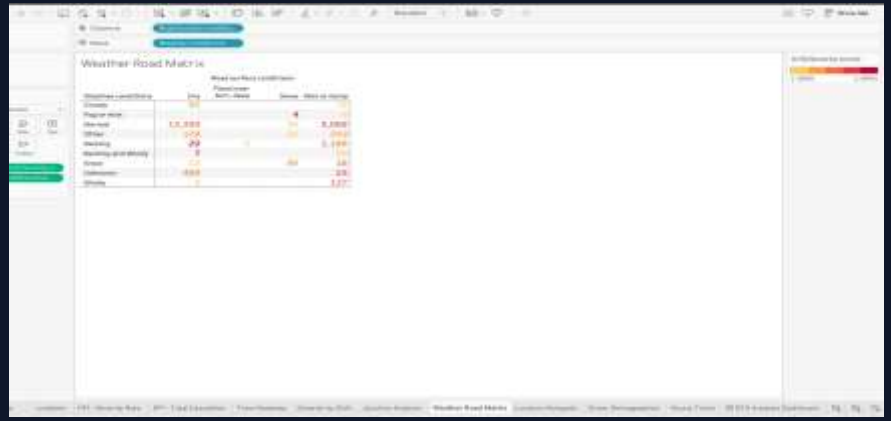


Driver Demographics (Bubbles)

Top 15 Causes (Treemap)

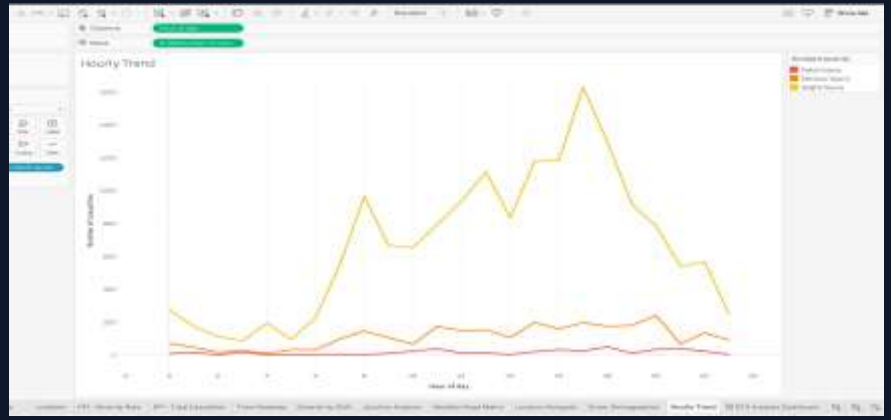
Dashboard — Additional Views

Junction Type Analysis



Weather × Road Matrix

Location Hotspots (Casualties)



Hourly Trend by Severity

Business Recommendations

1

Reallocate patrol resources: 50%+ to evening shift, surge at 15:00–19:00

EDA #1, #2 | Cover the 38.1% accident window that currently gets standard staffing — est. 15–20% reduction in peak-hour severe incidents

2

Launch "Safe Distance" enforcement campaign targeting tailgating

EDA #9 | Address the #1 cause (No distancing: 2,263 incidents, 18.4%) — direct enforcement has higher expected ROI than generic speed campaigns

3

Demographic-targeted awareness: male drivers aged 18–30

EDA #3, #8 | Reach the segment responsible for 34.7% of accidents + 92.9% male driver concentration — digital + campus outreach channels

4

Signal & visibility upgrades at Y-Shape and Crossing junctions

EDA #4 | Y-Shape alone accounts for 4,543 incidents — signal timing optimization and reflective signage are cost-effective interventions

5

Wet-weather early-warning system for arterial roads

Statistical #2 | Rain + dry roads (pre-wet phase) shows 1.29 avg severity — dynamic message signs alerting drivers at onset of rain in known hotspots

Limitations & Future Scope

Limitations

- No GPS/geolocation data — hotspot analysis relies on categorical area labels, not precise coordinates
- Single time-period dataset — no multi-year trend or seasonality analysis possible
- High 'Unknown' rate in some fields (e.g. 1,548 unknown age bands — 12.6%) limits demographic precision
- Severity_score is derived (1/2/3 ordinal scale), not externally validated against injury cost or recovery data
- Kaggle-sourced dataset may not represent a specific city — findings are indicative, not prescriptive for a named jurisdiction

Future Scope

- Integrate real-time GPS and traffic camera data for precise geospatial heatmapping
- Build predictive ML models (e.g. random forest on severity) to classify incoming incidents by risk tier
- Connect to live data streams for a real-time Tableau dashboard with automated alerting thresholds
- Expand to multi-city comparative analysis for benchmarking urban safety interventions
- Layer cost-of-accident financial modeling to enable ROI-based prioritization of infrastructure spend

Thank You

Team ModelCraft — Capstone 2

GitHub: github.com/Necro-Rohan/DVA-ModelCraft-Retail-Project

Institute: Newton School of Technology, Pune

Tableau: [Insert Tableau Public URL]

Date: April 2026